Muhammad Ameer Hamza

Senior Full-Stack Engineer | SaaS Architect | Production Automation Specialist

2 Live SaaS Products • 20+ Automation Clients • AWS • Python • MERN • Laravel • .NET

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PROFESSIONAL SUMMARY

Full-stack engineer and serial builder with production experience across SaaS development, enterprise automation, and AI-powered systems. Independently built and launched 2 revenue-generating SaaS platforms (FBR e-invoicing, bank statement converter deployed on AWS), operate production automation infrastructure serving 20+ MCA Global clients with 99.9%+ uptime, and delivered 3 enterprise platforms for hospitality and financial services firms. Expert across Python, MERN, Laravel, .NET 8, and distributed systems with hands-on experience in performance optimization (N+1 elimination, connection pooling, async), load/stress testing, AES-256 encryption, OAuth 2.0, and staged deployment pipelines. Currently building a pipeline of 6 additional SaaS applications across fintech, mobile, and business management verticals.

TECHNICAL SKILLS

Languages	JavaScript / TypeScript, Python, PHP, C#
Frontend	React.js, Next.js 14–16, React 18/19, Tailwind CSS, TanStack Query v5, Zustand
Backend	Node.js, NestJS, Express.js, Laravel, .NET 8 (ASP.NET Core), FastAPI, Flask
Databases	PostgreSQL, MongoDB, MySQL, SQLite, Prisma 7, Entity Framework Core, Redis
AI / ML	OpenAI GPT-4o API, XGBoost, LightGBM, Prophet, RandomForest, NLP, Scikit-learn
Automation / RPA	Python + Playwright, Gmail API, IMAP/SMTP, Windows Task Scheduler, Batch Scripting
DevOps / Cloud	AWS EC2, Docker, Nginx, Vercel, Render, GitHub Actions (CI/CD), Git
Security	JWT (access + refresh tokens), OAuth 2.0, AES-256 Encryption, RBAC, CSRF, HSTS
Testing	Load & Stress Testing, Edge Case Testing, UAT, Jest, Playwright E2E, pytest
Integrations	Stripe, FBR API, Open-Meteo, VAPID Push, Resend, PyMuPDF, Tesseract OCR, ClosedXML

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer | MCA Global (Main Course Associates)

07/2024 – Present

Islamabad, Pakistan • Enterprise Platforms + Production Automation Infrastructure

Production Email Automation Platform | 20+ Active Client Deployments

- Architect and operate production email automation infrastructure serving 15+ MCA Global clients with custom rule-based routing, Gmail label management, and client-specific business logic — invoices classified and forwarded, statements routed to dedicated labels, conditional CC/BCC per client.

- Built multi-tenant architecture: shared core Python engine with per-client directories containing isolated OAuth 2.0 credentials, config YAML files, and dedicated log directories — zero credential cross-contamination across clients.
- Achieved 99.9%+ uptime over 6+ months of production operation — zero critical failures; proactive log monitoring and root-cause analysis on rare incidents (OAuth token expiry) resolved before client impact.
- Deliver self-service deployment packages (.bat executables) for non-technical colleagues, with Windows Task Scheduler automation for fully hands-off operation.
- Execute formal UAT process: manual runs → maturation period → colleague sign-off → production scheduling. Change minimization philosophy prevents regression across isolated client codebases.
- Maintain daily AES-256 encrypted backups of all client directories including tokens, configs, and logs.

Technologies: *Python, Gmail API, OAuth 2.0, YAML Config Management, Windows Task Scheduler, AES-256, Structured Logging*

Production RPA Bot Fleet | 5–7 Vendor Portal Deployments

- Engineered and maintain production RPA bots automating invoice extraction from B2B vendor portals (Booker UK, FreshDirect, Castle Water) — completely eliminating manual document retrieval for financial operations.
- Built config-driven shared RPA codebase: single core with per-client config files and isolated OAuth tokens — new client onboarding requires only config + credential setup, minimizing duplication.
- Implemented idempotent SQLite-based processing pipeline preventing duplicate invoice downloads across concurrent client executions; integrated automated SMTP delivery confirmation.
- Perform root cause analysis using structured per-client logs; apply minor fixes in-place or rollback to previous codebase snapshot for significant changes.

Technologies: *Python, Playwright, SQLite, OAuth 2.0, SMTP/Gmail API, YAML Config, Structured Logging*

AI-Powered Financial Planning & Budgeting System | MCA Budgeting

- Engineered enterprise-grade multi-tenant financial planning platform for multi-site hospitality/retail clients — multi-stack architecture spanning Next.js 14 frontend, .NET 8 backend, Python FastAPI AI microservice, and PostgreSQL.
- Optimized system performance handling 2–3 years of multi-client data: eliminated N+1 query patterns, implemented PostgreSQL connection pooling (ThreadedConnectionPool 2–50 connections), and applied async processing — reducing load times from minutes to seconds.
- Built AI-powered budget generation with OpenAI GPT-4o integration, Prophet-based forecasting, seasonal adjustments, and financial validation; deployed on Vercel (frontend) and Render (backend).
- Developed automated weekly P&L flash reporting (JSON with parameterized breakdown: sales/COGS/payroll/costs), robust ETL pipeline supporting multiple Excel formats, and complex fiscal year handling with variance analysis.
- Implemented comprehensive testing suite: Jest unit tests, integration tests, Playwright E2E (multiple suites: dashboard, budgets, calculation accuracy, lifecycle), 20+ versioned SQL migrations.
- Executed gradual client migration strategy — phasing clients onto platform incrementally while maintaining backward compatibility; building login system for per-colleague client visibility.

Technologies: *Next.js 14, .NET 8, FastAPI, PostgreSQL, Prisma, Docker, OpenAI API, Prophet, ClosedXML, QuestPDF, Playwright*

Practice Management Platform | MCA Orbit

- Developed full-stack collaboration platform for financial services firms: Next.js 16, React 19, TypeScript, PostgreSQL, Prisma 7 — 20 database models, 23 RESTful APIs with sophisticated filtering and pagination.
- Built recurring task generation engine with frequency-based date arithmetic, idempotency checks, and 5-state lifecycle state machine with audit logging.
- Implemented real-time chat using Server-Sent Events (SSE) with exponential backoff reconnection, multi-thread subscriptions, and graceful disconnection handling.
- Designed multi-layer RBAC (4 system roles + 3 client roles), multi-channel notification system (email via Resend, web push via VAPID, in-app), and 20-field activity audit logging for compliance.
- Built compliance alert system for regulatory deadlines (VAT Returns, Corp Tax, Payroll) with customizable pre-due notifications and multi-user distribution.
- Engineered security hardening: bcrypt password hashing, CSRF protection, HSTS, X-Frame-Options, JWT with 8-hour session max and real-time active-status verification.

Technologies: *Next.js 16, React 19, TypeScript, PostgreSQL, Prisma 7, NextAuth.js, TanStack Query v5, Zustand, SSE, Resend, VAPID*

Founder & Full-Stack Developer | fbreinvoicing.com (Self-Founded SaaS) *2025 – Present*

Remote • Production SaaS • AWS EC2 Deployment

- Architected and launched production-ready multi-tenant FBR e-invoicing SaaS platform from scratch — owning full stack from UI and APIs to database design, security, and cloud deployment.

- Implemented real-time invoice submission to FBR API with IRN/USIN tracking, QR-code generation, and immutable invoice locking post-approval; backend deployed on AWS EC2 with whitelisted IP (FBR compliance requirement).
- Designed core PostgreSQL/Prisma data model for tenants, users, invoices, credit notes, audit logs, and FBR API logs with complete tenant isolation.
- Built secure authentication: JWT with Redis-backed refresh-token revocation, role-based access control, encrypted FBR credential storage (AES-256), and comprehensive API audit logging.
- Shipped Dockerized production setup: Next.js 14, NestJS, Redis, Nginx — reliable deployment with container orchestration.
- Pricing model: invoice-volume tiers (e.g., PKR 2,300/month for 10 invoices); platform currently in active marketing phase.

Technologies: *Next.js 14, NestJS, PostgreSQL, Prisma, Redis, Docker, Nginx, AWS EC2, JWT, AES-256, FBR API*

Full-Stack Developer | Bank Statement to Excel Converter SaaS (Self-Built) *05/2025 – Present*

Remote • Production SaaS • 100% Accuracy on Real-World Data

- Built production SaaS converting bank statement PDFs to structured Excel with 100% accuracy validated on 440 real transactions — supports digital and scanned documents via PyMuPDF, pdfplumber, and Tesseract OCR fallback.
- Engineered 5-tier subscription system with Stripe payment integration, JWT authentication (15-min access + 30-day refresh tokens), and usage tracking/rate limiting enforcing plan quotas.
- Developed Flask backend with 21 RESTful endpoints across authentication, file processing, payments, and developer API — background job processing, multi-format export (Excel/CSV/JSON), comprehensive error handling.
- Designed Next.js + TypeScript frontend: drag-and-drop upload, real-time conversion progress, paginated conversion history, subscription management, usage statistics, and developer API key management.
- Conducted thorough load testing and stress testing validating system stability under concurrent user load before production release.

Technologies: *Next.js, TypeScript, Flask, SQLite, Stripe, PyMuPDF, pdfplumber, Tesseract OCR, JWT, REST API*

Freelance Full-Stack Developer | Enterprise POS & Inventory Management System *03/2025 – Present*

Remote • Laravel • FBR API Integration

- Developed enterprise-grade Point of Sale and inventory management system using Laravel — secure authentication, advanced role-based policies, multi-page financial reporting with data-validation gates and self-healing cost-of-sales computations.
- Built asynchronous data pipelines using Laravel Jobs and Queues for bulk product imports and inventory restocks without blocking system performance.
- Engineered real-time FBR API tax compliance integration — automated invoice submission and audit logging.
- Established automated testing pipeline (Unit, Feature, E2E with Laravel Dusk) with comprehensive edge case coverage; conducted stress testing validating performance under extreme load.
- Implemented multi-page dynamic PDF generation for financial reports using industry-standard libraries.

Technologies: *Laravel, PHP, MySQL, Laravel Jobs/Queues, Laravel Dusk, FBR API, PDF Generation*

Freelance Full-Stack Developer | Client Projects & Inventory Systems *05/2025 – Present*

Remote • MERN Stack

- Built inventory management systems for multi-location textile and appliance businesses: real-time stock tracking, automated low-stock alerts, multi-location tracking, purchase/sale workflows, product-level P&L.
- Developed medical supplies inventory system with customer credit (Khata) management, dynamic pricing controls, and comprehensive reporting modules.
- Built CrossBorderLegal.pk with modern UI/UX, fully responsive layout, SEO optimization, and streamlined content structure for higher engagement.

Technologies: *React.js, Node.js, Express.js, MongoDB, JWT, REST APIs*

Full-Stack Web Developer Intern | National Information Technology Board (NITB) *07/2024 – 08/2024*

Islamabad, Pakistan • Government Digital Infrastructure

- Contributed to internal government dashboards and management tools; designed and optimized backend APIs with MongoDB data models; assisted in UI enhancements and application performance optimization.

NOTABLE PROJECTS

NBA Player Ranking System | Learning-to-Rank (LTR) with LightGBM

- Developed ranking system using LightGBM with custom feature engineering evaluating scoring, defense, and playmaking; automated per-36 statistics, efficiency metrics, and composite scoring for position-based dynamic queries.

Smart Disease Predictor | XGBoost ML Classifier

- Predicts diseases from symptom input using XGBoost classifier trained on medical datasets — telemedicine use case with probability estimation; extensible for severity estimation and prescription recommendations.

Task Tracker | MERN Application

- MERN stack app with JWT authentication, role-based access, attendance management, task assignment, deadline tracking, and real-time updates — responsive across desktop and mobile.

PRODUCTS IN PIPELINE

Serial builder with 6 additional products in development:

- **OnBoardly** — CRM + client onboarding platform with team chat, payments, and workflow management (in progress)
- **Goala** — AI-powered mobile goal achievement app: plan generation, daily task notifications, progress tracking (React Native, in progress)
- **ExpensBook** — React Native mobile app for business expense management targeting Pakistan SME market (in progress)
- **Advanced POS** — AI analytics, FBR integration, full inventory and financial reporting (in progress)
- **Tailor Management System** — Vertical SaaS for tailor shops (frontend complete)
- **Booking System** — Fresha-style appointment booking platform (frontend complete)

EDUCATION

BS Software Engineering | Air University, Islamabad *09/2021 – 07/2025*

F.Sc Pre-Engineering | The Quaid-e-Azam Group of Schools & Colleges, KP *09/2019 – 08/2021*

Matriculation (Science) | Army Public School and College, Zamzama *04/2016 – 11/2018*

CERTIFICATIONS & LANGUAGES

CERTIFICATIONS
Operating Systems — Coursera

LANGUAGES
English • Urdu • Pashto

KEY STRENGTHS

- **Production Operations Experience —** Operate 20+ live automation deployments with 99.9%+ uptime, log-driven incident response, AES-256 encrypted backups, and staged deployment pipelines.
- **Serial Builder & Founder —** Independently built 2 live revenue-generating SaaS platforms, 3 enterprise systems, and 6 products in pipeline — from idea to production without a team.
- **Full-Stack Versatility —** Proficient across MERN, Python, Laravel, and .NET 8; adapts stack to problem requirements rather than forcing one solution.
- **AI & Automation Depth —** Integrates ML/DL, NLP, RPA, and LLM APIs into production systems — not theoretical knowledge but shipped, client-facing features.
- **Security-First Engineering —** AES-256 encryption, JWT refresh-token revocation, OAuth 2.0 credential isolation, RBAC, and audit logging across all production systems.

- **Performance Engineering —** Proven optimization of N+1 queries, connection pooling, async processing, and load/stress testing before production deployment.